

KANAD S.H.I.E.L.D. 2026 · AHMEDABAD CITY POLICE · I-HUB GUJARAT

Paritran

FROM COMPLAINT TO CONVICTION

1.6%

the cybercrime conviction rate India is stuck at.
Paritran builds the court-ready case file that
detection tools never produce.

Team Paritran · Ayush Tiwary · Aditya Arora

PS-69EEFE4F8CD1C · Category 2 · Problem Statement 10 (Open-Ended Smart Policing)

THE PROBLEM

Detection is solved. Conviction is not.

₹22,845cr

LOST TO CYBER FRAUD, 2024

2.43%

NCRP REPORTS → FIRS

18%

CHARGESHEETING RATE

1.6%

CONVICTION RATE

**India's cyber-justice gap
18.1%**

2.43%

NCRP → FIR



Chargesheet

1.6%

Conviction

Three things turn volume into almost no justice



The money is gone before anyone acts

Fraud moves across mules in minutes; banks take ~6 days. ₹8,189 cr saved of ₹20,000 cr.



The case is never built to court standard

Hundreds of small complaints belong to a few syndicates, never linked; evidence lacks a clean custody chain.



The legal bar just went up

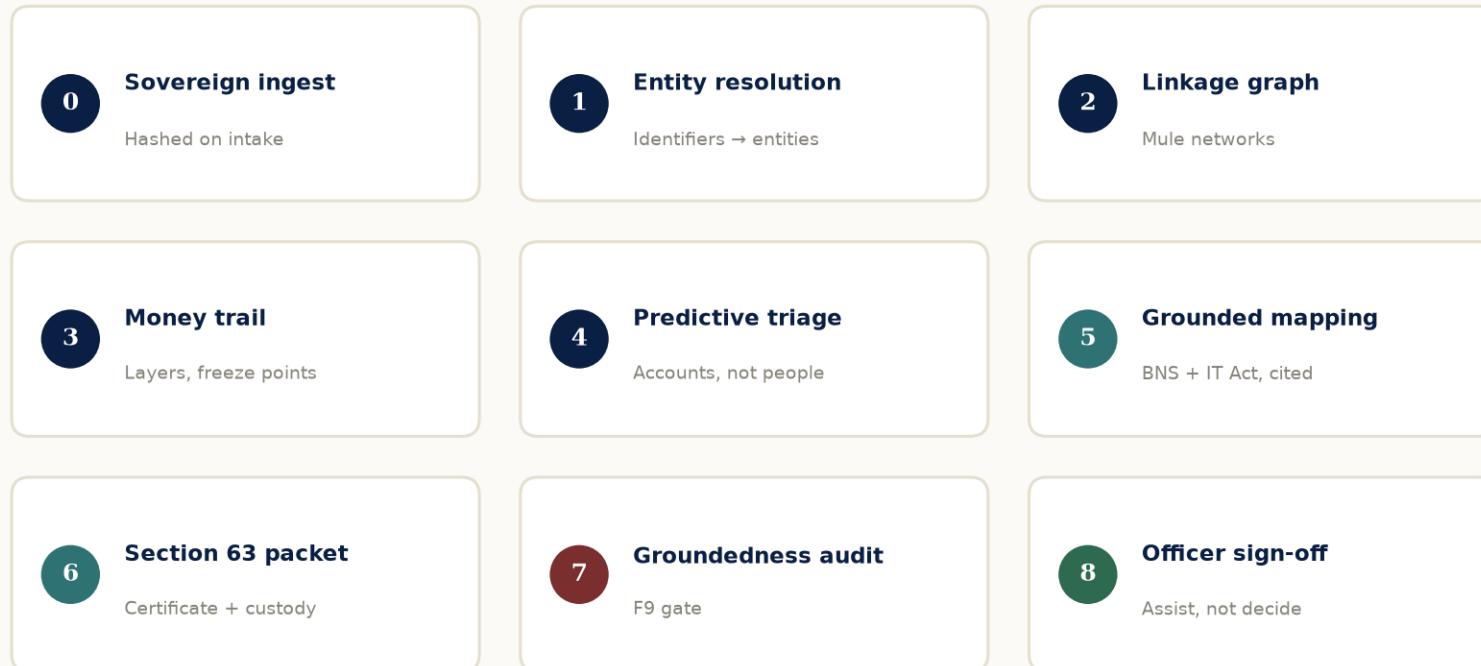
Since 1 July 2024, BNS, BNSS and BSA replaced IPC/CrPC/Evidence Act. Section 63 now needs dual certification.

Gujarat: about ₹678 cr from ~72,000 victims (Jan-Sep 2025). Ahmedabad logged the most cyber-fraud calls in the state, 28,382.

PROPOSED SOLUTION

The case file that holds up in court

Paritrان runs inside the branch. It links complaints into mule networks, rebuilds the money trail with a tamper-evident custody chain, drafts the Section 63 certificate, and maps every offence to the right law, quoted word for word. The officer reviews and signs. The engine never decides.



Why it wins

- Court-ready, not just detected**
We own the packet that ends in a conviction.
- Deterministic core, fenced model**
Same input, same output; the model only drafts, gated.
- On-premise, zero egress**
Victim data never leaves the branch. Works in Gujarati.
- Accounts, not people**
Scores money and infrastructure; no profiling.

SYSTEM ARCHITECTURE

One deterministic core, a fenced language layer



Ingest layer

Text + voice, multilingual · SHA-256 hash on intake · zero egress



Intelligence core (deterministic)

Entity resolution · linkage graph · money-trail reachability · triage



Grounded language layer (fenced)

BM25 + InLegalBERT + rules agree · model only drafts cited evidence



Assurance layer

F9 groundedness gate · SHA-256 hash-chained custody · officer sign-off



Deployment

On-premise / gov-cloud · Docker offline bundle · API-first CCTNS + ICJS

Holds up under scrutiny

- Offline-first, runs air-gapped in a station
- Deterministic core: reproducible at a fixed seed
- PII stays inside the branch boundary
- Model down → retrieval + rules + human review
- Hash-chain anchored out of band
- Concurrency-safe append-only audit ledger

TECHNOLOGY STACK

Built to run inside a police station

● Frontend

- React + TypeScript + Vite
- react-force-graph, Motion
- cmdk palette, react-countup
- TanStack Query, SSE live

● Backend + data

- Python 3.11, FastAPI
- PostgreSQL + RLS audit
- NetworkX graphs
- Plain-SQL migrations

● AI / ML

- InLegalBERT + BM25 retrieval
- Rule + retrieval agreement
- Gemma via Ollama (local)
- spaCy NER, Tesseract OCR

● Security + ops

- 7 scanners in CI
- OWASP ASVS 5.0
- Prometheus + Grafana
- Docker offline bundle

On-premise on purpose. Complaint and victim data never leave the branch, under DPDP and data-sovereignty norms. We treat that as a requirement and a strength.

DEMONSTRATION

Watch it work

Product views from the working prototype, filled with live seed-42 output. The chrome is designed; the numbers, graphs and hashes are real.

ParitransubmenuFrom complaint to conviction

Discovery

Demo

Case File

Custody

Evaluation

Security

SYSTEM STATUS

supervisor1

Discovery & Triage

SEED42

GENERATORstub (deterministic, labelled)

Start run

run seed 42 . generator stub . status completed

ingest

entity_resolution

linkage

money_trail

trriage

legal_mapping

f9_audit

packet

signoff

COMPLAINTS INGESTED

297

metric: n_complaints

NETWORKS FOUND

6

network.discovered events

RUPEES AT RISK

₹7,17,16,000

₹6,50,98,000 traced to cash-out (90.8%)

LINKAGE QUALITY

P 0.957

R 0.966

F1 0.962

appears when the linkage stage completes

297 nodes . 2287 links

LINK INSPECTOR

Click a link in the graph (or pick one below) to inspect it and, if it is wrong, reject it. Every rejection lands on the audit chain.

1 to 18 . W 3

121 to 127 . W 3

121 to 162 . W 3

126 to 156 . W 3

127 to 162 . W 3

TRIAGE QUEUE (RECOVERABILITY)

ParitransubmenuFrom complaint to conviction

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Case File

RUNseed 42 . stub . b90a72dea7d4d44b05febdcf32b1e9

NETWORKnetwork 0 . 56 complaints . syndicate 3

run b90a72dea7d4d44b05febdcf32b1e9 . seed 42 . generator stub

HONEY TRAIL (DIRECTED-GRAPH REACHABILITY)

VALUE TRACED TO CASH-OUT (RUN-WIDE)

90.8%

₹1,19,65,000

metric: pct_value_traced_to_cashout of ₹1,19,65,000 complaint value . 5 ledger hops . syndicate 3

VICTIMS

LAYER 1 MULES

LAYER 2 MULES

CASH-OUT

No break edges in this network at this seed: every ledger hop to cash-out is present, so there is no freeze point to flag here (₹1,19,65,000 of ₹1,19,65,000 traced). Run-wide, 90.8% of complaint value traced to cash-out; the untraced 9.2% is value the reachability walk never reached a cash-out account for (results totals), not a break in this network.

HAPPENED SECTIONS (VERBATIM CORPUS V2)

BNS 336 Forgery.

India Code accessed 2026-07-12

Act 45 of 2023, India Code digital text as served on 2026-07-12

BNS 111 Organised crime.

India Code accessed 2026-07-12

Act 45 of 2023, India Code digital text as served on 2026-07-12

BNS 308 Extortion.

India Code accessed 2026-07-12

Act 45 of 2023, India Code digital text

F9 GROUNDEDNESS AUDIT

GENERATORstub (deterministic, labelled)

Run F9 audit

generator deterministic-stub . corpus v1 . deterministic stub

PASSED

WITTFIELD

40

10

LEARNED

CLAIMS

0.962

LINKAGE F1

90.8%

VALUE TRACED

100%

HIGH-CONF MAPPING

40/10/0

F9 STUB GATE

verified

CUSTODY + TAMPER

INNOVATION

The two rows nobody else ticks

Capability	I4C / Pratibimb	RBI MuleHunter	StaQu CrimeGPT	Legal AI	Paritran
Fraud & mule detection	Yes	Yes	Search	No	Consumes
Court-ready Section 63 packet	No	No	No	No	YES
Tamper-evident custody	No	No	No	No	YES
Grounded mapping, no fabrication	N/A	N/A	Generative	Partial	F9 gated
Runs on-premise, zero egress	Gov cloud	Bank	Claimed	Cloud	Yes

Precision via abstention

100%

accuracy when three independent paths (rule, BM25, InLegalBERT + model) agree, on 8 of 8 and 15 of 15 golden cases.

Everything else routes to a human.

The moat: the Section 63 admissible packet and the tamper-evident custody chain. Everyone optimises detection. Paritran owns the part that ends in a conviction.

Why not just use ChatGPT? A police station needs offline, sovereign, legally-grounded, auditable, local-language output that supports admissibility. A public chatbot gives none of these.

IMPACT

Hours to seconds, money kept in reach



In parallel: golden-hour triage → higher freeze rate → more money recovered.



Public value: faster, lawful action for victims, and officer hours returned to investigation instead of paperwork.

SCALABILITY & FUTURE SCOPE

A path, and a companion

1 Phase 1 BUILT

Demo slice

Ingest, linkage, money trail, Section 63, audit gate. Built and measured.

2 Phase 2 NEXT

Branch pilot

Standalone on-premise, Gujarati drafting, officer UX. 90 days.

3 Phase 3 DESIGNED

Integrated

API-first with CCTNS, ICJS, CFCFRMS as interfaces open.

One core, two faces

The same grounded-mapping and audit engine also answers the crime-documentation statement. Standalone on-premise on day one.

Sustainable and procurable

Deployable via GeM and empanelment; hosting and maintenance costed for the branch. Winners get a deployment pathway with the Police.

SUPPORTING DOCUMENTS & RESOURCES

Verify everything in ten seconds

Fixed seed. Reproducible. Synthetic data only, zero real PII. Scan any code and run it yourself.



Source code / repository



Live website



Demo video



Solution document (PDF)



Research white paper



Reproduce results



Dataset datasheet



Contact Team Paritran

Ready to pilot with the Cyber Crime Branch

Paritran. From complaint to conviction.

Team Paritran · Ayush Tiwary · Aditya Arora

team@paritran.in · PS-69EEFE4F8CD1C · KANAD S.H.I.E.L.D. 2026

Thank you. Questions welcome.

Measured, reproducible, honest

0.962

LINKAGE F1

P 0.957 · R 0.966

90.8%

VALUE TRACED

graph reachability

100%

MAPPING, HIGH-CONFIDENCE

8/8 · 15/15, rest routed

0

F9 LEAKED

40/10/0 stub · live gated

0.74/1.0

NER P/R

seed 42

verified

CUSTODY + TAMPER

SHA-256 chain

Reproduce it on any seed you name

191 tests green · 34-second offline demo · seed 42 regenerates every number.

A judge types any seed in the Evaluation screen and every value moves; no canned number survives.

pip install networkx==3.4.2 && python3 src/paritran_prototype.py reproduces results.json standalone.

We corrected our own numbers

During prep we audited our own submitted appendix, found three numbers produced by placeholder code, replaced them with the real methods, re-measured, and published both columns in a public changelog. Our numbers went from claimed to reproducible on any seed.

	WHAT WE SUBMITTED		WHAT WE NOW RUN
Money trail	87.3% random draw	→	90.8% graph reachability
Section mapping	90.5% keyword ladder	→	52.4% floor · 100% high-confidence
F9 groundedness	0/50 tautology	→	gated over live model, 0 leaked
Linkage F1	0.936	→	0.962 re-measured

Honesty is the deliverable. The one genuine vulnerability, owned in the open, becomes the strongest trust signal in the room.

Security you can see, not just claims

Paritrán

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SYSTEM STATUS

db ok

ollama ok

model_files ok

latency: no request data yet

supervisor1

SUPERVISOR

logout

ran Jul 12, 2026, 09:14:09 PM

pip-audit reports no severity field; findings_total only

ran Jul 12, 2026, 09:14:10 PM

npm 'moderate' counted as medium; 'info' as low

ran Jul 12, 2026, 09:14:10 PM

bandit has no 'critical' grade; high is its maximum

ran Jul 12, 2026, 09:14:14 PM

semgrep ERROR->high, WARNING->medium, INFO->low

ran Jul 12, 2026, 09:14:15 PM

every detected secret counted as critical

TRIVY-FS

0 CRITICAL · 1 HIGH

medium 1 · low 2 · unknown 0 · total 4

ran Jul 12, 2026, 09:14:15 PM

TRIVY-IMAGE

3 CRITICAL · 42 HIGH

medium 82 · low 131 · unknown 30 · total 288

ran Jul 12, 2026, 09:14:16 PM

GRYPE

0 CRITICAL · 1 HIGH

medium 1 · low 2 · unknown 0 · total 4

ran Jul 12, 2026, 09:14:17 PM

EGRESS: MEASURED, NOT ASSERTED

Live self-test: outbound TCP to 1.1.1.1:443 is open (checked Jul 13, 2026, 07:09:12 PM, 2s timeout).

TCP connect succeeded; this process CAN reach the internet right now. At the venue the demo runs with Wi-Fi off, which this same test then shows as blocked, live. Docker cannot both publish ports and block WAN on the same network, so zero egress is proven by this live attempt from inside the api process, not asserted from config.

CONFIGURED OUTBOUND ENDPOINTS (THE COMPLETE LIST)

http://host.docker.internal:11434 . generative step for the F9-gated language layer; host-local model gemma3:4b, the only configured HTTP egress

postgresql://db:5432 . Postgres DSN host (credentials omitted); on the internal compose network in the container deployment

OWASP TOP 10:2025 COVERAGE

Each category mapped to the concrete control in this repo; "partial" names the gap instead of hiding it. Full table with the ASVS 5.0 alignment and accepted risks: docs/SECURITY.md.

100% Broken Access Control

Observability + privacy

- Prometheus + Grafana, 8-panel dashboard
- OpenTelemetry traced, custom metrics live
- Live system-status widget over SSE
- DPDP-aligned, on-premise, zero egress
- Encryption at rest (host FDE), TLS for pilot
- Hash-chain head anchored out of band

One button, five beats, offline

1 Intake

297 complaints stream in; counters climb, ₹ at risk rises.

2 Collapse

Force graph resolves 297 into 6 mule networks; officer rejects one link (hashed).

3 Money trail

Value flows victim → cash-out; traced % climbs to 90.8.

4 Packet + F9

Section 63 assembles with verbatim quotes; a planted fabrication is blocked live.

5 Custody

Hash chain renders; tamper test breaks a scratch chain, real chain stays verified.

Runs with Wi-Fi off, the zero-egress proof, live. Fallbacks: captioned video and a stub-degrade banner, all numbers stay live.

Reproduce standalone: `python3 src/paritran_prototype.py`

Evidence integrity, and reproduce it yourself

Paritrان

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auditor1

AUDITOR

logout

seq 4 **tamper_test.run** by auditor1 Jul 13, 2026, 07:00:47 PM

prev 4f22a5d2b2..f6a248 hash 53219c8eb3..17fb8e

{"break_seq":2,"scratch_rows":3,"corrupted_seq":2,"real_chain_ok":true}

seq 5 **decision.link.reject** by supervisor1 Jul 13, 2026, 07:07:16 PM

prev 53219c8eb3..17fb8e hash 255de9d66d..19b8c8

{"ref":{"a":1,"b":10,"w":3},"kind":"link","role":"supervisor","run_id":"18190051ee784de98837d787d..

seq 6 **decision.link.reject** by supervisor1 Jul 13, 2026, 07:07:16 PM

prev 255de9d66d..19b8c8 hash 5cbd1bbb8b..5f333c

{"ref":{"a":121,"b":127,"w":3},"kind":"link","role":"supervisor","run_id":"18190051ee784de98837d..

seq 7 **decision.link.reject** by supervisor1 Jul 13, 2026, 07:07:17 PM CORRUPTED IN SCRATCH COPY . CHAIN BREAKS HERE

prev 5cbd1bbb8b..5f333c hash 87cb5f6549..0849e1

{"ref":{"a":121,"b":162,"w":3},"kind":"link","role":"supervisor","run_id":"18190051ee784de98837d..

seq 8 **decision.link.reject** by supervisor1 Jul 13, 2026, 07:07:18 PM

prev 87cb5f6549..0849e1 hash 858675b580..eb6e14

{"ref":{"a":126,"b":156,"w":3},"kind":"link","role":"supervisor","run_id":"18190051ee784de98837d..

seq 9 **decision.link.reject** by supervisor1 Jul 13, 2026, 07:07:19 PM

prev 858675b580..eb6e14 hash 5dd837e85c..2abae4

{"ref":{"a":127,"b":162,"w":3},"kind":"link","role":"supervisor","run_id":"18190051ee784de98837d..

seq 10 **decision.link.reject** by supervisor1 Jul 13, 2026, 07:07:19 PM

prev 5dd837e85c..2abae4 hash 5b5969a1e6..90c964

{"ref":{"a":128,"b":143,"w":3},"kind":"link","role":"supervisor","run_id":"18190051ee784de98837d..

Paritrان

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db ok

ollama ok

model_files ok

p50 / p95 request latency, last 10 ticks (max 19 ms)

supervisor1

SUPERVISOR

logout

REPRODUCE THE SEED-42 BASELINE (SUPERVISOR)

Reproduce

✓ Every compared key matches the committed baseline exactly. The numbers regenerated live.

KEY	BASELINE (RESULTS.JSON)	FRESH RUN	MATCH
n_complaints	297	297	✓ equal
n_syndicates_seeded	6	6	✓ equal
networks_found	6	6	✓ equal
linkage_precision	0.957	0.957	✓ equal
linkage_recall	0.966	0.966	✓ equal
linkage_f1	0.962	0.962	✓ equal
pct_value_traced_to_cashout	90.8	90.8	✓ equal
money_trail_method	directed-graph reachability	directed-graph reachability	✓ equal
section_accuracy_bm25	52.4	52.4	✓ equal
section_method	Okapi BM25 over condensed section-description corpus (v1)	Okapi BM25 over condensed section-description corpus (v1)	✓ equal
f9_claims	50	50	✓ equal
f9_passed	40	40	✓ equal
f9_withheld_stub_fabrications	10	10	✓ equal
f9_leaked	0	0	✓ equal
chain_len	12	12	✓ equal
chain_verified	true	true	✓ equal
tamper_detected	true	true	✓ equal
time_to_packet_sec	0.045	15.29	live wall clock, not compared
data	synthetic, ground-truth known.	synthetic, ground-truth known.	✓ equal

The custody chain is tamper-evident and the whole pipeline reproduces at a fixed seed. A judge can break a record and watch it flag, or type any seed and watch every number move.